MIL-M-38510/3F
20 MAY 1985
SUPERSEDING
MIL-M-0038510/3E(19)
1 December 1975 and
MIL-M-38510/3B
23 December 1971
(See 6.6)

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, NAND BUFFERS

MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic, silicon, TTL, positive NAND buffer microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Dual, 4-input, positive NAND buffer
02	Quadruple, 2-input positive NAND buffer
03	Quadruple, 2-input positive NAND buffer (open collector output)

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
A	F-1, (14-pin, 1/4" x 1/4"), flat package
B	F-3, (14-pin, 3/16" x 1/4"), flat package
C	D-1, (14-pin, 1/4" x 3/4"), dual-in-line package
D	F-2, (14-pin, 1/4" x 3/8"), flat package

1.2.4 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.5 V dc at -12 mA to +5.5 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) 1/	100 mW dc per buffer
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}) - Cases A, B, C, and D 3/	(see MIL-M-38510, appendix C)
Junction temperature (T_J) 2/	+175°C

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).

2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5004 of MIL-STD-883.

3/ When a thermal resistance value is included in MIL-M-38510, appendix C, it shall supersede the value stated herein.

Beneficial comments (recommendations, additions, deletions) and pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5962

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1.2.5 Recommended operating conditions.

Supply voltage (V_{CC})- - - - - - - - - - - -	4.5 V dc minimum to 5.5 V dc maximum
Minimum high-level input voltage (V_{IH})- -	2.0 V dc
Maximum low-level input voltage (V_{IL}) - -	0.8 V dc
Normalized fanout (each output) 4/- - - -	30 maximum
Case operating temperature range (T_C) - -	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and herein.

3.2.1 Logic diagrams and terminal connections. The logic diagram and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers schematics shall be maintained by the agent activity and will be available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range.

4/ The device shall fanout in both high and low levels to the specified number of inputs of the same device type as that being tested.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55^{\circ}C \leq T_C \leq +125^{\circ}C$	Device type	Limits Min	Limits Max	Units
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IN} = 0.8 V I_{OH} = -1.2 mA 1/	01,02	2.4		V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 48 mA V_{IN} = 2.0 V for all inputs of gate under test 1/	01,02,03		0.4	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -12 mA T_C = +25°C	01,02,03		-1.5	V
High-level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V 2/	01,02,03		40	µA
High-level input current	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V 2/	01,02,03		100	µA
Low-level input current	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.4 V 1/	01,02,03	-0.7	-1.6	mA
Short-circuit output current	I_{OS}	V_{CC} = 5.5 V 2/ 3/	01,02	-20	-70	mA
Maximum collector cut-off-current	I_{CEX}	V_{OH} = 5.5 V V_{CC} = 4.5 V, V_{IN} = 0.8 V	03		250	µA
High-level supply current per buffer	I_{CCH}	V_{CC} = 5.5 V 2/, V_{IN} = 0 V	01		8	mA
			02		15.5	mA
			03		8.5	mA
Low-level supply current per buffer	I_{CCL}	V_{CC} = 5.5 V 1/ V_{IN} = 5.5 V	01		27	mA
			02		54	mA
			03		54	mA
Propagation delay time, high-to-low-level	t_{PHL}	C_L = 125 pF ±10% R_L = 133Ω ±5%	01,02,03	3	30	ns
Propagation delay time, low to high-level	t_{PLH}	C_L = 125 pF ±10% R_L = 133Ω ±5%	01,02,03	3	27	ns

1/ All unspecified inputs at 5.5 volts.

2/ All unspecified inputs grounded.

3/ Not more than one output should be shorted at a time.

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3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3, 9	1*,2,3, 9
Group A test requirements (method 5005)	1,2,3, 9,10,11	1,2,3, 9
Group B test requirements (method 5005) subgroup 5	1,2,3, 9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3
Additional electrical subgroups for group C periodic inspections	N/A	10,11

*PDA applies to subgroup 1 (see 4.3c).

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 2 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1, through 4.4.3).

4.3 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

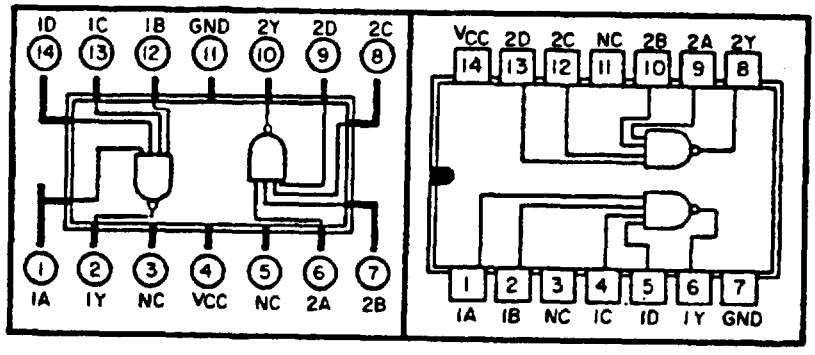
a. Burn-in (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 3, or equivalent.

(2) T_C = +125°C minimum.

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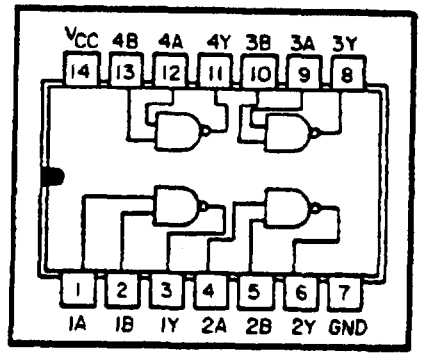
Device type 01



Case A, B and D **Case C**

NC-No Internal Connection

Device types 02 and 03



Cases A, B, C, and D

FIGURE 1. Logic diagrams and terminal connections (top views).

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Device type 01

Truth table				
Input				Output
A	B	C	D	Y
L	L	L	L	H
H	L	L	L	H
L	H	L	L	H
H	H	L	L	H
L	L	H	L	H
H	L	H	L	H
L	H	H	L	H
H	H	H	L	H
L	L	L	H	H
H	L	L	H	H
L	H	L	H	H
H	H	L	H	H
L	L	H	H	H
H	L	H	H	H
L	H	H	H	H
H	H	H	H	L

Positive logic $Y = \overline{ABCD}$

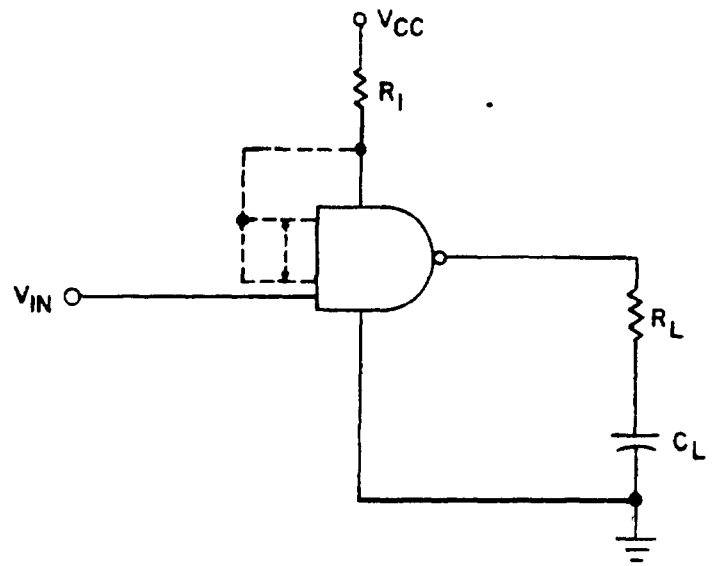
Device types 02 and 03

Truth table each gate		
INPUT		OUTPUT
A	B	Y
L	L	H
H	L	H
L	H	H
H	H	L

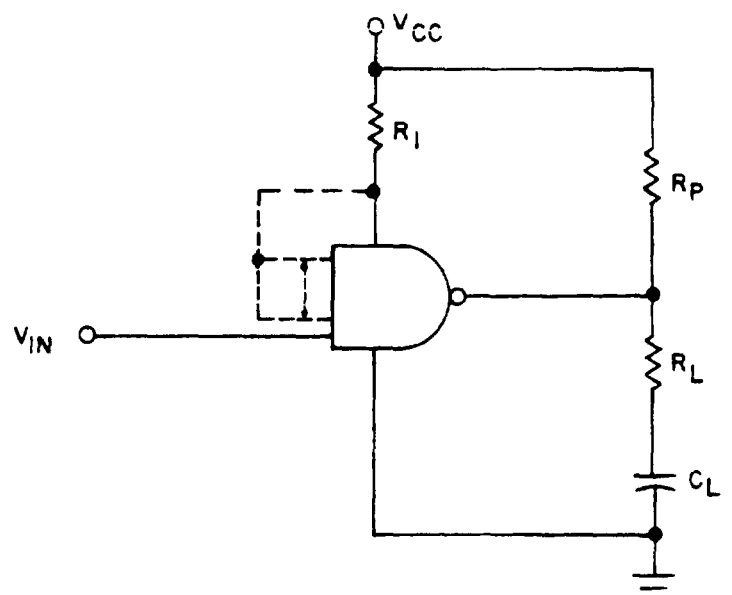
Positive Logic $Y = \overline{AB}$

FIGURE 2. Truth tables.

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CIRCUIT A



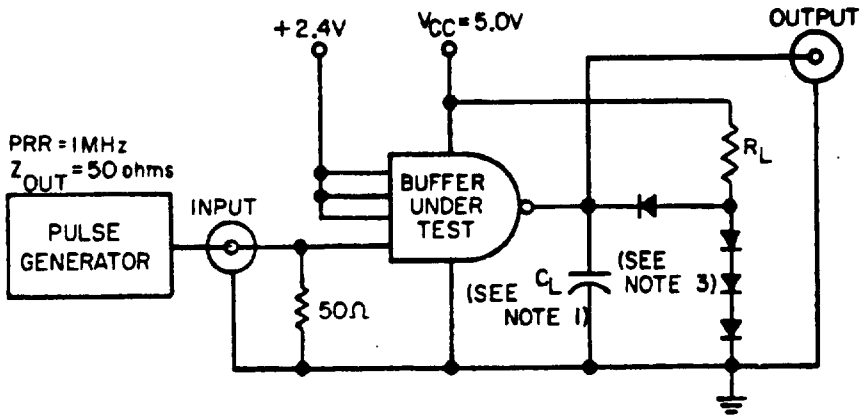
CIRCUIT B

NOTES:
1. One input of each buffer shall be driven. The remaining inputs of each buffer shall be connected to V_{CC} through the appropriate current limiting resistors.
2. V_{CC} and R_1 are such that the minimum voltage at the device terminal is 5 volts.
3. If parallel excitation is used, the clock frequency shall be 100 kHz with V_{IN} = 3 volts minimum and minimum duty cycle of 50%.
4. R_L = 51Ω ±5%; C_L = .1μf±10%; R_p = 100Ω ±5%.
5. Circuit A shall be used for device types 01 and 02. Circuit B shall be used for device type 03.

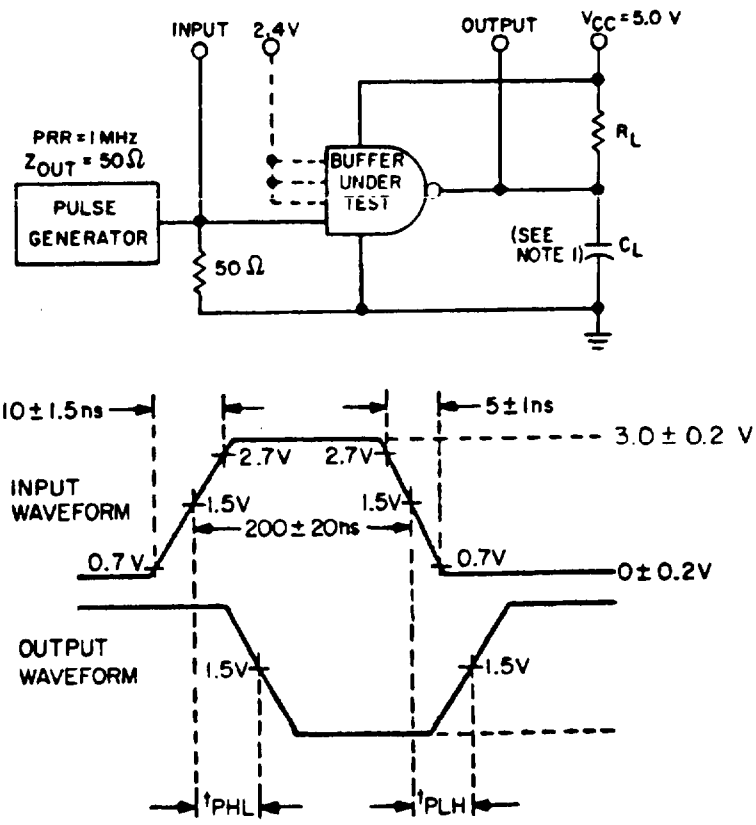
FIGURE 3. Burn-in and life test circuits.

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Device types 01 and 02



Device type 03



NOTES:
1. Including scope probe, wiring, and stray capacitance, without package in test fixture, C_L = 125 pF ±10%.
2. Voltage measurements are to be made with respect to network ground terminal.
3. All diodes are 1N3064 or equivalent.
4. R_L = 133Ω ±5%.

FIGURE 4. Switching time test circuit.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,D Case C Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits Min	Limits Max	Unit
				1	6	3	14	11	9	10	12	13	8	7	2	4	5				
				1A	1Y	NC	V_{CC}	NC	2A	2B	2C	2D	2Y	GND	1B	1C	1D				
1 T_C = +25°C	V_{OL}	3007	1	2.0 V	48 mA		4.5 V		5.5 V	5.5 V	5.5 V	5.5 V	48 mA	GND	2.0 V	2.0 V	2.0 V	1Y		0.4	V
		3007	2	5.5 V			"		2.0 V	2.0 V	2.0 V	2.0 V		"	5.5 V	5.5 V	5.5 V	2Y		0.4	"
	V_{OH}	3006	3	0.8 V	-1.2 mA		"		5.5 V	5.5 V	5.5 V	5.5 V		"	5.5 V	"	"	1Y	2.4		"
		"	4	5.5 V	"		"		"	"	"	"		"	0.8 V	"	"	1Y	"		"
		"	5	"	"		"		"	"	"	"		"	5.5 V	0.8 V	0.8 V	1Y	"		"
		"	6	"	"		"		"	"	"	"		"	"	5.5 V	5.5 V	2Y	"		"
		"	7	"	"		"		0.8 V	0.8 V	0.8 V	"	-1.2 mA	"	"	"	"	2Y	"		"
		"	8	"	"		"		5.5 V	5.5 V	5.5 V	0.8 V	"	"	"	"	"	2Y	"		"
		"	9	"	"		"		"	"	"	5.5 V	"	"	"	"	"	2Y	"		"
		"	10	"	"		"		"	"	"	"	"	"	"	"	"	2Y	"		"
	V_{IC}		11	-12 mA			"		-12 mA					"	-12 mA			1A		-1.5	"
			12				"			-12 mA				"		-12 mA		1B		"	"
			13				"				-12 mA			"			-12 mA	1C		"	"
			14				"					-12 mA		"				1D		"	"
			15				"							"				2A		"	"
			16				"							"				2B		"	"
			17				"							"				2C		"	"
			18				"							"				2D		"	"
	I_{OS}	3011	19	GND	GND		5.5 V		GND	GND	GND	GND	GND	"	GND	GND	GND	1Y	-20	-70	mA
		3011	20	"	"		"		"	"	"	"	"	"	"	"	"	2Y	-20	-70	mA
	I_{IHL}	3010	21	2.4 V			"		"	"	"	"		"	GND	GND	GND	1A		40	µA
		"	22	GND			"		"	"	"	"		"	2.4 V	2.4 V	"	1B		"	"
		"	23	"			"		"	"	"	"		"	GND	GND	2.4 V	1C		"	"
		"	24	"			"		"	"	"	"		"	"	"	GND	1D		"	"
		"	25	"			"		"	"	"	"		"	"	"	"	2A		"	"
		"	26	"			"		"	"	"	"		"	"	"	"	2B		"	"
		"	27	"			"		"	"	"	"		"	"	"	"	2C		"	"
		"	28	"			"		2.4 V	2.4 V	2.4 V	2.4 V		"	"	"	"	2D		"	"
									GND	GND	GND										
	I_{IH2}	"	29	5.5 V			"		"	"	"	"		"	"	"	"	1A		100	"
		"	30	GND			"		"	"	"	GND		"	5.5 V	"	"	1B		"	"
		"	31	"			"		"	"	"	"		"	GND	5.5 V	"	1C		"	"
		"	32	"			"		"	"	"	"		"	"	GND	5.5 V	1D		"	"
		"	33	"			"		"	"	"	"		"	"	"	GND	2A		"	"
		"	34	"			"		"	"	"	"		"	"	"	"	2B		"	"
		"	35	"			"		"	"	"	"		"	"	"	"	2C		"	"
		"	36	"			"		5.5 V	5.5 V	5.5 V	5.5 V		"	"	"	"	2D		"	"
									GND	GND	GND										
	I_{IL}	3010	37	0.4 V			"		5.5 V	5.5 V	5.5 V	"		"	5.5 V	5.5 V	5.5 V	1A	-0.7	-1.6	mA
		"	38	5.5 V			"		"	"	"	"		"	0.4 V	5.5 V	"	1B	"	"	"
		"	39	"			"		"	"	"	"		"	5.5 V	0.4 V	5.5 V	1C	"	"	"
		"	40	"			"		"	"	"	"		"	"	"	"	1D	"	"	"
		"	41	"			"		0.4 V	0.4 V	0.4 V	"		"	"	"	0.4 V	2A	"	"	"
		"	42	"			"		5.5 V	5.5 V	5.5 V	"		"	"	"	5.5 V	2B	"	"	"
		"	43	"			"		"	"	"	"		"	"	"	"	2C	"	"	"
		"	44	"			"		"	"	"	0.4 V		"	"	"	"	2D	"	"	"
											0.4 V	5.5 V									
											5.5 V	0.4 V									
	I_{CCL}	"	45	"			"		"	"	5.5 V	5.5 V		"	"	"	"	V_{CC}		27	"
	I_{CCH}	"	46	GND			"		GND	GND	GND	GND		"	GND	GND	GND	V_{CC}		8	"

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,D / Case C / Test no.	1 / 1 / IA	2 / 6 / 1Y	3 / 3 / NC	4 / 14 / V_{CC}	5 / 11 / NC	6 / 9 / 2A	7 / 10 / 2B	8 / 12 / 2C	9 / 13 / 2D	10 / 8 / 2Y	11 / 7 / GND	12 / 2 / 1B	13 / 4 / 1C	14 / 5 / 1D	Measured terminal	Limits Min	Limits Max	Unit
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
9 T_C = +25°C	t_{PHL}	3003 (Fig. 4)	47	IN	OUT		5.0 V		IN	2.4 V	2.4 V	2.4 V	OUT	GND	2.4 V	2.4 V	2.4 V	1A to 1Y	3	25	ns
	t_{PHL}	"	48				"							"				2A to 2Y	"	25	"
	t_{PLH}	"	49	IN	OUT		"		IN	2.4 V	2.4 V	2.4 V	OUT	"	2.4 V	2.4 V	2.4 V	1A to 1Y	"	22	"
	t_{PLH}	"	50				"							"				2A to 2Y	"	22	"
10 T_C = +125°C	t_{PHL}	3003 (Fig. 4)	51	IN	OUT		"		IN	2.4 V	2.4 V	2.4 V	OUT	"	2.4 V	2.4 V	2.4 V	1A to 1Y	"	30	"
	t_{PHL}	"	52				"							"				2A to 2Y	"	30	"
	t_{PLH}	"	53	IN	OUT		"		IN	2.4 V	2.4 V	2.4 V	OUT	"	2.4 V	2.4 V	2.4 V	1A to 1Y	"	27	"
	t_{PLH}	"	54				"							"				2A to 2Y	"	27	"
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																				

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case C	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	
			Test no.	1A	1B	1Y	2A	2B	2Y	GND	3Y	3A	3B	4Y	4A	4B	V_{CC}				
1 T_C = +25°C	V_{OL}	3007	1	2.0 V	2.0 V	48 mA	5.5 V	5.5 V		GND		5.5 V	5.5 V		5.5 V	5.5 V	4.5 V	1Y		0.4	V
		"	2	5.5 V	5.5 V		2.0 V	2.0 V	48 mA	"		5.5 V	5.5 V		"	"	"	2Y		"	"
		"	3	"	"		5.5 V	5.5 V		"	48 mA	2.0 V	2.0 V		"	"	"	3Y		"	"
		"	4	"	"		"	"		"		5.5 V	5.5 V	48 mA	2.0 V	2.0 V	"	4Y		"	"
	V_{OH}	3006	5	0.8 V	"	-1.2 mA	"	"		"		"	"		5.5 V	5.5 V	"	1Y	2.4		"
		"	6	5.5 V	0.8 V	-1.2 mA	"	"		"		"	"		"	"	"	1Y	"		"
		"	7	"	5.5 V		0.8 V	"	-1.2 mA	"		"	"		"	"	"	2Y	"		"
		"	8	"	"		5.5 V	0.8 V	-1.2 mA	"		"	"		"	"	"	2Y	"		"
		"	9	"	"		"	5.5 V	"	"	-1.2 mA	0.8 V	"		"	"	"	3Y	"		"
		"	10	"	"		"	"		"	-1.2 mA	5.5 V	0.8 V		"	"	"	3Y	"		"
		"	11	"	"		"	"		"		"	5.5 V	-1.2 mA	0.8 V	"	"	4Y	"		"
		"	12	"	"		"	"		"		"	5.5 V	-1.2 mA	5.5 V	0.8 V	"	4Y	"		"
	V_{IC}		13	-12 mA						"							"	1A		-1.5	"
			14		-12 mA					"							"	1B		"	"
			15				-12 mA			"							"	2A		"	"
			16					-12 mA		"							"	2B		"	"
			17							"		-12 mA					"	3A		"	"
			18							"			-12 mA				"	3B		"	"
			19							"					-12 mA		"	4A		"	"
			20							"						-12 mA	"	4B		"	"
	I_{OS}	3011	21	GND	GND	GND				"							5.5 V	1Y	-20	-70	mA
		"	22				GND	GND	GND	"	GND	GND	GND		GND	GND	"	2Y	"	"	"
		"	23							"				GND			"	3Y	"	"	"
		"	24							"							"	4Y	"	"	"
	I_{IH1}	3010	25	2.4 V	GND		GND	GND		"		GND	GND		"	"	"	1A		40	μA
		"	26	GND	2.4 V		GND	"		"		"	"		"	"	"	1B		"	"
		"	27	"	GND		2.4 V	"		"		"	"		"	"	"	2A		"	"
		"	28	"	"		GND	2.4 V		"		"	"		"	"	"	2B		"	"
		"	29	"	"		"	GND		"		2.4 V	"		"	"	"	3A		"	"
		"	30	"	"		"	"		"		GND	2.4 V		"	"	"	3B		"	"
		"	31	"	"		"	"		"		"	GND		2.4 V	"	"	4A		"	"
		"	32	"	"		"	"		"		"	GND		GND	2.4 V	"	4B		"	"
	I_{IH2}	"	33	5.5 V	"		"	"		"		"	"		"	GND	"	1A		100	"
		"	34	GND	5.5 V		"	"		"		"	"		"	"	"	1B		"	"
		"	35	"	GND		5.5 V	"		"		"	"		"	"	"	2A		"	"
		"	36	"	"		GND	5.5 V		"		"	"		"	"	"	2B		"	"
		"	37	"	"		"	GND		"		5.5 V	"		"	"	"	3A		"	"
		"	38	"	"		"	"		"		GND	5.5 V		"	"	"	3B		"	"
		"	39	"	"		"	"		"		"	GND		5.5 V	"	"	4A		"	"
		"	40	"	"		"	"		"		"	GND		GND	5.5 V	"	4B		"	"
	I_{IL}	"	41	0.4 V	5.5 V		5.5 V	5.5 V		"		5.5 V	5.5 V		5.5 V	"	"	1A	-0.7	-1.6	mA
		"	42	5.5 V	0.4 V		5.5 V	"		"		"	"		"	"	"	1B	"	"	"
		"	43	"	5.5 V		0.4 V	"		"		"	"		"	"	"	2A	"	"	"
		"	44	"	"		5.5 V	0.4 V		"		"	"		"	"	"	2B	"	"	"
		"	45	"	"		"	5.5 V		"		0.4 V	"		"	"	"	3A	"	"	"
		"	46	"	"		"	"		"		5.5 V	0.4 V		"	"	"	3B	"	"	"
		"	47	"	"		"	"		"		"	5.5 V		0.4 V	"	"	4A	"	"	"
		"	48	"	"		"	"		"		"	5.5 V		5.5 V	0.4 V	"	4B	"	"	"
	I_{CCL}	"	49	5.0 V	5.0 V		5.0 V	5.0 V		"		5.0 V	5.0 V		5.0 V	5.0 V	"	V_{CC}		54	"
	I_{CCH}	"	50	GND	GND		GND	GND		"		GND	GND		GND	GND	"	V_{CC}		15.5	"

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case C	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	
			Test no.	1A	1B	1Y	2A	2B	2Y	GND	3Y	3A	3B	4Y	4A	4B	V_{CC}				
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
9 T_C = +25°C	t_{PHL}	3003 (Fig. 4)	51	IN	2.4 V	OUT				GND							5.0 V	1A to 1Y	3	25	ns
		"	52				IN	2.4 V	OUT	"	OUT	IN	2.4 V				"	2A to 2Y	"	"	"
		"	53							"				OUT	IN	2.4 V	"	3A to 3Y	"	"	"
		"	54							"							"	4A to 4Y	"	"	"
	t_{PLH}	3003 (Fig. 4)	55	IN	2.4 V	OUT				"							"	1A to 1Y	"	22	"
		"	56				IN	2.4 V	OUT	"	OUT	IN	2.4 V				"	2A to 2Y	"	"	"
		"	57							"				OUT	IN	2.4 V	"	3A to 3Y	"	"	"
		"	58							"							"	4A to 4Y	"	"	"
10 T_C = +125°C	t_{PHL}	3003 (Fig. 4)	59	IN	2.4 V	OUT				"							"	1A to 1Y	"	30	"
		"	60				IN	2.4 V	OUT	"	OUT	IN	2.4 V				"	2A to 2Y	"	"	"
		"	61							"				OUT	IN	2.4 V	"	3A to 3Y	"	"	"
		"	62							"							"	4A to 4Y	"	"	"
	t_{PLH}	3003 (Fig. 4)	63	IN	2.4 V	OUT				"							"	1A to 1Y	"	27	"
		"	64				IN	2.4 V	OUT	"	OUT	IN	2.4 V				"	2A to 2Y	"	"	"
		"	65							"				OUT	IN	2.4 V	"	3A to 3Y	"	"	"
		"	66							"							"	4A to 4Y	"	"	"
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																				

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TABLE III. Group A inspection for device type 03.

Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,D Case C Test no.	1 1A	2 1B	3 1Y	4 2A	5 2B	6 2Y	7 GND	8 3Y	9 3A	10 3B	11 4Y	12 4B	13 4B	14 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	V_{OL}	3007	1	2.0 V	2.0 V	48 mA	5.5 V	5.5 V	48 mA	GND		5.5 V	5.5 V		5.5 V	5.5 V	4.5 V	1Y		0.4	V
		"	2	5.5 V	5.5 V		2.0 V	2.0 V		"		5.5 V	5.5 V		"	"	"	2Y		"	"
		"	3	"	"		5.5 V	5.5 V		"	48 mA	2.0 V	2.0 V		2.0 V	2.0 V	"	3Y		"	"
		"	4	"	"		"	"		"		5.5 V	5.5 V	48 mA			"	4Y		"	"
	I_{CEX}	"	5	0.8 V	"	5.5 V	"	"		"		"	"		5.5 V	5.5 V	"	1Y		250	μA
		"	6	5.5 V	0.8 V	5.5 V		"	5.5 V	"		"	"		"	"	"	1Y		"	"
		"	7	"	5.5 V	"	0.8 V	"	5.5 V	"		"	"		"	"	"	2Y		"	"
		"	8	"	"	"	5.5 V	0.8 V	"	"		0.8 V	"		"	"	"	2Y		"	"
		"	9	"	"	"	"	5.5 V	"	"	5.5 V	5.5 V	0.8 V		"	"	"	3Y		"	"
		"	10	"	"	"	"	"	"	"	5.5 V	"	5.5 V	5.5 V	0.8 V	"	"	3Y		"	"
		"	11	"	"	"	"	"	"	"		"	5.5 V	5.5 V	5.5 V	0.8 V	"	4Y		"	"
		"	12	"	"	"	"	"	"	"		"	"		"	"	"	4Y		"	"
	V_{IC}	"	13	-12 mA						"								1A		-1.5	V
		"	14		-12 mA					"								1B		"	"
		"	15				-12 mA			"								2A		"	"
		"	16					-12 mA		"								2B		"	"
		"	17							"		-12 mA						3A		"	"
		"	18							"			-12 mA					3B		"	"
		"	19							"					-12 mA			4A		"	"
		"	20							"						-12 mA		4B		"	"
	I_{IH1}	3010	21	2.4 V	GND		GND	GND		"		GND	GND		GND	GND	5.5 V	1A		40	μA
		"	22	GND	2.4 V		GND	"		"		"	"		"	"	"	1B		"	"
		"	23	"	GND		2.4 V	"		"		"	"		"	"	"	2A		"	"
		"	24	"	"		GND	2.4 V		"		"	"		"	"	"	2B		"	"
		"	25	"	"		"	GND		"		2.4 V	"		"	"	"	3A		"	"
		"	26	"	"		"	"		"		GND	2.4 V		"	"	"	3B		"	"
		"	27	"	"		"	"		"		"	GND		2.4 V	"	"	4A		"	"
		"	28	"	"		"	"		"		"	"		GND	2.4 V	"	4B		"	"
	I_{IH2}	"	29	5.5 V	"		"	"		"		"	"		"	"	"	1A		100	"
		"	30	GND	5.5 V		"	"		"		"	"		"	"	"	1B		"	"
		"	31	"	GND		5.5 V	"		"		"	"		"	"	"	2A		"	"
		"	32	"	"		GND	5.5 V		"		"	"		"	"	"	2B		"	"
		"	33	"	"		"	GND		"		5.5 V	"		"	"	"	3A		"	"
		"	34	"	"		"	"		"		GND	5.5 V		"	"	"	3B		"	"
		"	35	"	"		"	"		"		"	GND		5.5 V	"	"	4A		"	"
		"	36	"	"		"	"		"		"	"		GND	5.5 V	"	4B		"	"
	I_{IL}	3009	37	0.4 V	5.5 V		5.5 V	5.5 V		"		5.5 V	5.5 V		5.5 V	"	"	1A	-0.7	-1.6	mA
		"	38	5.5 V	0.4 V		5.5 V	"		"		"	"		"	"	"	1B	"	"	"
		"	39	"	5.5 V		0.4 V	"		"		"	"		"	"	"	2A	"	"	"
		"	40	"	"		5.5 V	0.4 V		"		"	"		"	"	"	2B	"	"	"
		"	41	"	"		"	5.5 V		"		0.4 V	"		"	"	"	3A	"	"	"
		"	42	"	"		"	"		"		5.5 V	0.4 V		"	"	"	3B	"	"	"
		"	43	"	"		"	"		"		"	5.5 V		0.4 V	"	"	4A	"	"	"
		"	44	"	"		"	"		"		"	"		5.5 V	0.4 V	"	4B	"	"	"
	I_{CCL}	3005	45	"	"		"	"		"		"	"		"	5.5 V	"	V_{CC}		54	"
	I_{CCH}	3005	46	GND	GND		GND"	GND		"		GND	GND		GND	GND	"	V_{CC}		9.5	"

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,D Case C Test no.	1 1 1A	2 2 1B	3 3 1Y	4 4 2A	5 5 2B	6 6 2Y	7 7 GND	8 9 3Y	9 9 3A	10 10 3B	11 11 4Y	12 12 4A	13 13 4B	14 14 V_{CC}	Measured terminal	Limits Min	Max	Unit
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
9 T_C = +25°C	t_{PHL}	3003 (Fig. 4)	47	IN	2.4 V	OUT	IN	2.4 V	OUT	GND	OUT	IN	2.4 V	OUT	IN	2.4 V	5.0 V	1A to 1Y	3	25	ns
		"	48							"							"	2A to 2Y	"	"	"
		"	49							"							"	3A to 3Y	"	"	"
		"	50							"							"	4A to 4Y	"	"	"
	t_{PLH}	3003 (Fig. 4)	51	IN	2.4 V	OUT	IN	2.4 V	OUT	"	OUT	IN	2.4 V	OUT	IN	2.4 V	"	1A to 1Y	"	22	"
		"	52							"							"	2A to 2Y	"	"	"
		"	53							"							"	3A to 3Y	"	"	"
		"	54							"							"	4A to 4Y	"	"	"
10 T_C = +125°C	t_{PHL}	3003 (Fig. 4)	55	IN	2.4 V	OUT	IN	2.4 V	OUT	GND	OUT	IN	2.4 V	OUT	IN	2.4 V	"	1A to 1Y	"	30	"
		"	56							"							"	2A to 2Y	"	"	"
		"	57							"							"	3A to 3Y	"	"	"
		"	58							"							"	4A to 4Y	"	"	"
	t_{PLH}	3003 (Fig. 4)	59	IN	2.4 V	OUT	IN	2.4 V	OUT	"	OUT	IN	2.4 V	OUT	IN	2.4 V	"	1A to 1Y	"	27	"
		"	60							"							"	2A to 2Y	"	"	"
		"	61							"							"	3A to 3Y	"	"	"
		"	62							"							"	4A to 4Y	"	"	"
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																				

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b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Test requirements shall be as specified in table II.

b. Subgroups 4, 5, 6, 7, and 8 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical parameters shall be as specified in table II herein.

4.4.3 Groups C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E, using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C, minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for use for government microcircuit applications (original equipment) and logistic purposes.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

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d. Requirements for notification of change of product or process to contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - - - - - - - Ground zero voltage potential.
V_{IN} - - - - - - - - - - - - - - - - - - Voltage level at an input terminal.
V_{IC} - - - - - - - - - - - - - - - - - - Input clamp voltage.

6.4 Logistic support. Lead materials and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-referenced information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Device type	Commercial type
01	5440
02	5437
03	5438

6.6 In-lieu of documents. MIL-M-0038510/3E(19) was issued as an in-lieu-of document for MIL-M-38510/3B. This revision, MIL-M-38510/3F dated 20 MAY 1985 supersedes MIL-M-38510/3C dated 15 January 1985, MIL-M-38510/3B dated 23 December 1971 and MIL-M-0038510/3E(19) dated 1 December 1975.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/3F	MICROCIRCUITS,DIGITAL,TTL,NAND BUFFERS MONOLITHIC SILICON

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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